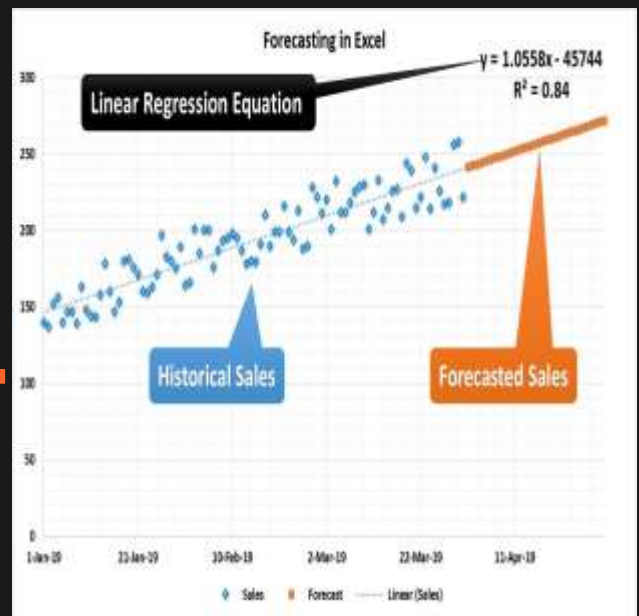


JAL2521

TIME SERIES ANALYSIS AND FORECASTING

UNIT III REGRESSION ANALYSIS AND FORECASTING



UNIT III REGRESSION ANALYSIS AND FORECASTING

SYLLABUS

Introduction - Least Squares Estimation in Linear Regression Models - Statistical Inference in Linear Regression - Prediction of New Observations - Model Adequacy Checking - Variable Selection Methods in Regression - Generalized and Weighted Least Squares-Regression Models for General Time Series Data.

Introduction

- Regression analysis → a statistical technique for modeling and investigating the relationships between an outcome or response variable and one or more predictor or regressor variables → a statistical method that can be used to predict future outcomes by analyzing historical data.
- The end result of a regression analysis study → to generate a model that can be used to forecast or predict future values of the response variable, given specified values of the predictor variables.

Introduction

- Simple linear regression model $y = \beta_0 + \beta_1 x + \epsilon$
- Here y → the response, x → the predictor variable, β_0 and β_1 → model parameters (or) regression coefficients – intercept & slope, and ϵ → an error term.
- The model parameters must be estimated from a sample of data → using the method of **least squares**.
- If there are k predictors, the multiple linear regression model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + \epsilon$
- Here $\beta_0, \beta_1, \dots, \beta_k$ → partial regression coefficients.
- Quadratic linear regression model $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon$

Introduction

- A linear regression model that describes the relationship between a response variable y that varies cyclically with time

$$y_t = \beta_0 + \beta_1 \sin \frac{2\pi}{d}t + \beta_2 \cos \frac{2\pi}{d}t + \varepsilon_t$$

- The nature of this cyclic variation can be described as a simple sine wave.
- The regression model for cross-section data (data collected over some specified study period (such as a month) and the data do not explicitly vary with time) for n individuals:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

- The regression model for time series data (data from same individual collected at different times – eg. hourly CO_2 emissions from a chemical plant)

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \dots + \beta_k x_{tk} + \varepsilon_t, \quad t = 1, 2, \dots, T$$

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

Regression model is used with cross-section data.

- $y_i \rightarrow$ n observations on the response variable $\rightarrow y_1, y_2, \dots, y_n$ (eg. n persons)
- $x_{ij} \rightarrow$ denotes represents the i^{th} observation of the j^{th} regressor or predictor variable (eg. Jan, Feb, Mar, ... data of i^{th} person)
- Assumption: Error term ε has $E(\varepsilon) = 0$ and variance $\text{Var}(\varepsilon) = \sigma^2$
- The method of least squares chooses the model parameters (the β 's) in the following Eq. so that the sum of the squares of the errors, ε_i , is minimized.

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

TABLE 3.1 Cross-Section Data for Multiple Linear Regression

Observation	Response, y	x_1	x_2	...	x_k
1	y_1	x_{11}	x_{12}	...	x_{1k}
2	y_2	x_{21}	x_{22}	...	x_{2k}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	y_n	x_{n1}	x_{n2}	...	x_{nk}

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

This function is to be minimized with respect to $\beta_0, \beta_1, \dots, \beta_k$.

Therefore the least squares estimators, say, $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$, must satisfy the following

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

$$L = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2} - \dots - \beta_k x_{ik})^2$$

$$= \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^k \beta_j x_{ij} \right)^2$$

$$\left. \frac{\partial L}{\partial \beta_0} \right|_{\beta_0, \beta_1, \dots, \beta_k} = -2 \sum_{i=1}^n \left(y_i - \hat{\beta}_0 - \sum_{j=1}^k \hat{\beta}_j x_{ij} \right) = 0$$

Eq. 1

$$\left. \frac{\partial L}{\partial \beta_j} \right|_{\beta_0, \beta_1, \dots, \beta_k} = -2 \sum_{i=1}^n \left(y_i - \hat{\beta}_0 - \sum_{j=1}^k \hat{\beta}_j x_{ij} \right) x_{ij} = 0, \quad j = 1, 2, \dots, k$$

Eq. 2

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

- Simplifying Eq.2,

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_{i1} + \hat{\beta}_2 \sum_{i=1}^n x_{i2} + \dots + \hat{\beta}_k \sum_{i=1}^n x_{ik} = \sum_{i=1}^n y_i$$

- Expanding for each Individual,

$$\begin{aligned} \hat{\beta}_0 \sum_{i=1}^n x_{i1} + \hat{\beta}_1 \sum_{i=1}^n x_{i1}^2 + \hat{\beta}_2 \sum_{i=1}^n x_{i2}x_{i1} + \dots + \hat{\beta}_k \sum_{i=1}^n x_{ik}x_{i1} &= \sum_{i=1}^n y_i x_{i1} \\ \vdots \\ \hat{\beta}_0 \sum_{i=1}^n x_{ik} + \hat{\beta}_1 \sum_{i=1}^n x_{i1}x_{ik} + \hat{\beta}_2 \sum_{i=1}^n x_{i2}x_{ik} + \dots + \hat{\beta}_k \sum_{i=1}^n x_{ik}^2 &= \sum_{i=1}^n y_i x_{ik} \end{aligned}$$

- These equations are called the least squares normal equations $\rightarrow$ has $p=k+1$ eqns..
- The solutions to the normal equations will be the least squares estimators of the model regression coefficients

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

- These equations can be rewritten as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1k} \\ 1 & x_{21} & x_{22} & \dots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nk} \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \text{and} \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

- Here $\mathbf{X} \rightarrow$ called as model matrix $\rightarrow$ the original data table for the problem expanded to the form of the regression model.

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

- The vector of least squares estimators (β) minimizes L.

$$L = \sum_{i=1}^n \varepsilon_i^2 = \varepsilon' \varepsilon = (y - X\beta)'(y - X\beta)$$

- Expanding the right-hand side of L, we obtain

$$L = y'y - \beta'X'y - y'X\beta + \beta'X'X\beta = y'y - 2\beta'X'y + \beta'X'X\beta$$

because $\beta'X'y$ is a (1x1) matrix

- The least square estimators (β) must satisfy the following:

$$\left. \frac{\partial L}{\partial \beta} \right|_{\hat{\beta}} = -2X'y + 2(X'X)\hat{\beta} = 0$$

- Simplified as

$$(X'X)\hat{\beta} = X'y$$

$$\hat{\beta} = (X'X)^{-1}X'y$$

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

- Thus the equation to estimate the model parameters is

$$\hat{\beta} = (X'X)^{-1}X'y$$

- And the equation to estimate or forecast the response variable is

$$\hat{y} = X\hat{\beta}$$

- The n residuals (The difference between the actual observation y_i and the corresponding fitted value $\rightarrow$ the residual) can be written as an $(n \times 1)$ vector denoted by

$$e = y - \hat{y} = y - X\hat{\beta}$$

- Sum of squares of the residuals

$$SS_E = (y - X\hat{\beta})'(y - X\hat{\beta})$$

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

• Example problem: Patient Satisfaction Survey

A hospital is implementing a program to improve quality and productivity. As part of this program, the hospital is attempting to measure and evaluate patient satisfaction. Table contains some of the data that have been collected for a random sample of 25 recently discharged patients. The “severity” variable is an index that measures the severity of the patient’s illness, measured on an increasing scale (i.e., more severe illnesses have higher values of the index), and the response satisfaction is also measured on an increasing scale, with larger values indicating greater satisfaction.

Observation	Age (x_1)	Severity (x_2)	Satisfaction (y)
1	55	50	68
2	46	24	77
3	30	46	96
4	35	48	80
5	59	58	43
6	61	60	44
7	74	65	26
8	38	42	88
9	27	42	75
10	51	50	57
11	53	38	56
12	41	30	88
13	37	31	88
14	24	34	102
15	42	30	88
16	50	48	70
17	58	61	52
18	60	71	43
19	62	62	46
20	68	38	56
21	70	41	59
22	79	66	26
23	63	31	52
24	39	42	83
25	49	40	75

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

Solution:

- Fit a multiple linear regression model to the patient satisfaction data.
- The model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

- To solve $\hat{\beta} = (X'X)^{-1}X'y$
- The model matrix X and observation vector y:
- The $X'X$ matrix and the $X'y$ vector:

$$X'X = \begin{bmatrix} 1 & 1 & \dots & 1 \\ 55 & 46 & \dots & 49 \\ 50 & 24 & \dots & 40 \end{bmatrix} \begin{bmatrix} 1 & 55 & 50 \\ 1 & 46 & 24 \\ \vdots & \vdots & \vdots \\ 1 & 49 & 40 \end{bmatrix} = \begin{bmatrix} 25 & 1271 & 1148 \\ 1271 & 69881 & 60814 \\ 1148 & 60814 & 56790 \end{bmatrix}$$

$$X'y = \begin{bmatrix} 1 & 1 & \dots & 1 \\ 55 & 46 & \dots & 49 \\ 50 & 24 & \dots & 40 \end{bmatrix} \begin{bmatrix} 68 \\ 77 \\ \vdots \\ 75 \end{bmatrix} = \begin{bmatrix} 1638 \\ 76487 \\ 70426 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 55 & 50 \\ 1 & 46 & 24 \\ 1 & 30 & 46 \\ 1 & 35 & 48 \\ 1 & 59 & 58 \\ 1 & 61 & 60 \\ 1 & 74 & 65 \\ 1 & 38 & 42 \\ 1 & 27 & 42 \\ 1 & 51 & 50 \\ 1 & 53 & 38 \\ 1 & 41 & 30 \\ 1 & 37 & 31 \\ 1 & 24 & 34 \\ 1 & 42 & 30 \\ 1 & 50 & 48 \\ 1 & 58 & 61 \\ 1 & 60 & 71 \\ 1 & 62 & 62 \\ 1 & 68 & 38 \\ 1 & 70 & 41 \\ 1 & 79 & 66 \\ 1 & 63 & 31 \\ 1 & 39 & 42 \\ 1 & 49 & 40 \end{bmatrix} \quad y = \begin{bmatrix} 68 \\ 77 \\ 96 \\ 80 \\ 43 \\ 44 \\ 26 \\ 88 \\ 75 \\ 57 \\ 56 \\ 88 \\ 88 \\ 102 \\ 88 \\ 70 \\ 52 \\ 43 \\ 46 \\ 56 \\ 59 \\ 26 \\ 52 \\ 83 \\ 75 \end{bmatrix}$$

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

Solution:

- Find the least squares estimates of the parameters in the regression model as below: $\hat{\beta} = (X'X)^{-1}X'y$

- Therefore the regression model is $\hat{\beta} = (X'X)^{-1}X'y$

$$\hat{y} = 143.472 - 1.031x_1 - 0.556x_2$$

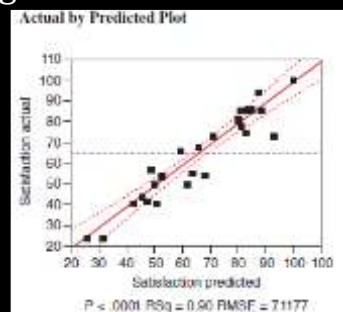
where x_1 = patient age and
 x_2 = severity of illness

$$\begin{aligned} &= \begin{bmatrix} 25 & 1271 & 1148 \\ 1271 & 69881 & 60814 \\ 1148 & 60814 & 56790 \end{bmatrix}^{-1} \begin{bmatrix} 1638 \\ 76487 \\ 70426 \end{bmatrix} \\ &= \begin{bmatrix} 0.699946097 & -0.006128086 & -0.007586982 \\ -0.006128086 & 0.00026383 & -0.000158646 \\ -0.007586982 & -0.000158646 & 0.000340866 \end{bmatrix} \begin{bmatrix} 1638 \\ 76487 \\ 70426 \end{bmatrix} \\ &= \begin{bmatrix} 143.4720118 \\ -1.031053414 \\ -0.55603781 \end{bmatrix} \end{aligned}$$

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

Solution Analysis:

- The points seem to scatter closely along the 45° line, suggesting that the model is a reasonably good fit to the data.
- If the fit is “perfect” then the actual-predicted and the plotted points would lie on a straight 45° line



LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

Solution Analysis:

- The residuals e can be calculated and tabulated

$$e = y - \hat{y} = y - X\hat{\beta}$$

Observation	Age	Severity	Satisfaction	Residual
1	55	50	68	0.03781647
2	46	24	77	-5.6986473
3	30	46	96	0.03732088
4	35	48	80	-0.6953274
5	39	58	43	-7.3896674
6	61	60	44	-3.2154849
7	74	65	26	-5.0316015
8	38	42	88	7.06160995
9	27	42	75	-12.279962
10	51	50	57	-6.0863972
11	53	38	56	-11.696764
12	41	30	88	3.48231247
13	37	31	88	-0.0858634
14	24	34	102	2.17855567
15	42	30	88	4.51336588
16	50	48	70	4.73047378
17	58	61	52	2.24739262
18	60	71	43	0.86987755
19	62	62	46	0.92764409
20	68	38	56	3.76905713
21	70	41	59	10.4992774
22	79	66	26	0.67970337
23	63	31	52	-9.2784746
24	39	42	83	3.09265936
25	49	80	75	4.29111788

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

Example-2 – Trend Adjustment:

- A time series that exhibits a trend is a nonstationary time series.
- Modeling and forecasting of such a time series is greatly simplified if we can eliminate the trend.
- One way to do this is to fit a regression model describing the trend component to the data and then subtracting it out of the original observations, leaving a set of residuals that are free of trend.
- The trend models that are usually considered are the linear trend, in which the mean of y_t is expected to change linearly with time as in

$$E(y_t) = \beta_0 + \beta_1 t$$

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

Example-2 – Trend Adjustment - Procedure:

- Fit a model with a linear trend term in time
- Subtract the fitted values from the original observations to obtain a set of residuals that are trend-free
- Forecast the residuals
- Compute the forecast → the forecast of the residual value(s) + estimate of trend.

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

Example-2 – Trend Adjustment - Procedure:

- The least squares normal equations for this model are

$$\begin{aligned} T\beta_0 + \beta_1 \sum_{t=1}^T t &= \sum_{t=1}^T y_t \\ \beta_0 \sum_{t=1}^T t + \beta_1 \sum_{t=1}^T t^2 &= \sum_{t=1}^T ty_t \end{aligned} \quad \longrightarrow \quad \begin{aligned} T\hat{\beta}_0 + \hat{\beta}_1 \frac{T(T+1)}{2} &= \sum_{t=1}^T y_t \\ \hat{\beta}_0 \frac{T(T+1)}{2} + \hat{\beta}_1 \frac{T(T+1)(2T+1)}{6} &= \sum_{t=1}^T ty_t \end{aligned}$$

- Because there are only two parameters, solving the normal equations directly, results in the least squares estimators

$$\begin{aligned} \hat{\beta}_0 &= \frac{2(2T+1)}{T(T-1)} \sum_{t=1}^T y_t - \frac{6}{T(T-1)} \sum_{t=1}^T ty_t \\ \hat{\beta}_1 &= \frac{12}{T(T^2-1)} \sum_{t=1}^T ty_t - \frac{6}{T(T-1)} \sum_{t=1}^T y_t \end{aligned}$$

LEAST SQUARES ESTIMATION IN LINEAR REGRESSION MODELS

Example-2 – Trend Adjustment - Procedure:

- The least squares estimates depend on T
- The estimates are denoted as functions of time, say, $\hat{\beta}_0(T)$ and $\hat{\beta}_1(T)$.
- The model can be used to predict the next observation at time T+1 as $\hat{\beta}_0(T) + \hat{\beta}_1(T)(T+1) + \hat{e}_{T+1}(1)$
- If the residuals have average value zero,

$$\hat{y}_{T+1}(T) = \hat{\beta}_0(T) + \hat{\beta}_1(T)(T + 1)$$

STATISTICAL INFERENCE IN LINEAR REGRESSION

- Test for Significance of Regression
- Tests on Individual Regression Coefficients and Groups of Coefficients
- Confidence Intervals on Individual Regression Coefficients
- Confidence Intervals on the Mean Response

STATISTICAL INFERENCE IN LINEAR REGRESSION

- In linear regression problems, certain tests of hypotheses about the model parameters and confidence interval estimates of these parameters are helpful in measuring the usefulness of the model.
- These procedures require that the errors ε_i in the model are normally and independently distributed with mean zero and variance σ^2 , abbreviated as NID(0, σ^2).

Test for Significance of Regression

- To determine whether there is a linear relationship between the response variable y and a subset of the predictor or regressor variables $x_1, x_2, \dots, x_k$.
- The appropriate hypotheses:

$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$ $H_1 : \text{at least one } \beta_j \neq 0$
--
- Null hypothesis $H_0 \rightarrow$ The regression model is not significant, meaning that the independent variable(s) have no effect on the dependent variable.
- Alternative hypothesis $H_1 \rightarrow$ at least one independent variable significantly contributes to predicting the dependent variable.

Test for Significance of Regression

- The total sum of squares
- It is partitioned into a sum of squares due to the model (or regression) and a sum of squares due to residual (or error),

$$SS_T = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$SS_T = SS_R + SS_E$$

- 2 approaches (or) test statistics used for testing:
 1. F-value approach
 2. P-value approach

Test for Significance of Regression

F-value approach

- If null hypothesis H_0 is TRUE then F_0 = Mean square regression (MSR) / Mean square error (MSE)

$$F_0 = \frac{SS_R/k}{SS_E/(n-p)}$$

- $F_0 \rightarrow$ the test statistic for significance of regression
- $k, n-p \rightarrow$ degrees of freedom, where $n \rightarrow$ no. of observations & $p \rightarrow$ no. of predictors
- H_0 is rejected if F_0 exceeds upper tail point of the F distribution

P-value approach

- H_0 is rejected if P-value for $F_0 < \alpha$ (significance level, typically 0.05 or 0.01)

Test for Significance of Regression

- The test for significance of regression is summarized in an analysis of variance (ANOVA) table

TABLE 3.4 Analysis of Variance for Testing Significance of Regression

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	Test Statistic, F_0
Regression	SS_R	k	$\frac{SS_R}{k}$	$F_0 = \frac{SS_R/k}{SS_E/(n-p)}$
Residual (error)	SS_E	$n - p$	$\frac{SS_E}{n - p}$	
Total	SS_T	$n - 1$		

- Computational formulas for the sums of squares in the ANOVA are

$$SS_T = \sum_{i=1}^n (y_i - \bar{y})^2 = y'y - n\bar{y}^2$$

$$SS_R = \hat{\beta}' X'y - n\bar{y}^2$$

$$SS_E = y'y - \hat{\beta}' X'y$$

Test for Significance of Regression

- The statistic $R^2 \rightarrow$ a measure of the amount of reduction in the variability of y obtained by using the predictor variables $x_1, x_2, \dots, x_k$ in the model $\rightarrow$ a measure of how well the regression model fits the data sample.

$$R^2 = \frac{SS_R}{SS_T} = 1 - \frac{SS_E}{SS_T}$$

- The adjusted $R^2 \rightarrow$ large value of the adjusted R^2 statistic are considered good regression models $\rightarrow$ with minimized residual mean square.

$$R_{Adj}^2 = 1 - \frac{SS_E/(n-p)}{SS_T/(n-1)}$$

Tests on Individual Regression Coefficients and Groups of Coefficients

Tests on Individual Regression Coefficients

- To determine the value or contribution of each predictor variable in the regression model.
- The inclusion of additional variables or the deletion of one or more of the variables already in the model might make the model more effective
- Adding a variable to the regression model → causes an increase in the sum of squares and decrease in the error sum of squares for regression → the usage of additional variable must be decided with justification.
- Adding an unimportant variable to the model → causes an increase in the MSE → causes a decrease in the usefulness of the model.
- The hypotheses for testing the significance of any individual regression coefficient β_j :

$$H_0 : \beta_j = 0$$

$$H_1 : \beta_j \neq 0$$

Tests on Individual Regression Coefficients and Groups of Coefficients

Tests on Individual Regression Coefficients

- If the null hypothesis $H_0 : \beta_j = 0$ is not rejected → indicates that the predictor variable x_j can be deleted from the model → x_j has no effect on the model.
- The test statistic for this hypothesis is **t-test**
- Where C_{jj} is the diagonal element of the $(X'X)^{-1}$ matrix corresponding to the regression coefficient β_j .
- Null hypothesis H_0 is rejected if $|t_0| > t_{\alpha/2, n-p}$, where $n-p$ → degree of freedom.
- If the standard error of the regression coefficient is se ,

$$H_0 : \beta_j = 0$$

$$H_1 : \beta_j \neq 0$$

$$t_0 = \frac{\hat{\beta}_j}{\sqrt{\hat{\sigma}^2 C_{jj}}}$$

$$se(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 C_{jj}}$$

$$t_0 = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}$$

Tests on Individual Regression Coefficients and Groups of Coefficients

Tests on Groups of Coefficients

- To test the contribution of a subset involving several regressor or predictor variables to the model.
- The least squares equation is $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \text{and} \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

- $\mathbf{y}$ is $(n \times 1)$, $\mathbf{X}$ is $(n \times p)$, $\boldsymbol{\beta}$ is $(p \times 1)$, $\boldsymbol{\varepsilon}$ is $(n \times 1)$, and $p = k + 1$, $k \rightarrow$ no. of variables, $n \rightarrow$ no. of individuals.
- It has to be determined if a subset of the predictor variables $x_1, x_2, \dots, x_r$, $r < k$, contributes significantly to the regression model.

Tests on Individual Regression Coefficients and Groups of Coefficients

Tests on Groups of Coefficients

- Consider $\boldsymbol{\beta} = \begin{bmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \end{bmatrix}$, where $\boldsymbol{\beta}_1$ is $(r \times 1)$ and $\boldsymbol{\beta}_2$ is $((p-r) \times 1)$.
- The full model may be rewritten as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$
- The hypotheses for testing is
$$\begin{matrix} H_0 : \boldsymbol{\beta}_1 = \mathbf{0} \\ H_1 : \boldsymbol{\beta}_1 \neq \mathbf{0} \end{matrix}$$
- Using the null hypothesis H_0 , the reduced model is
$$\mathbf{y} = \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$$
- The least squares estimator of $\boldsymbol{\beta}_2$
$$\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}_2' \mathbf{X}_2)^{-1} \mathbf{X}_2' \mathbf{y}$$

Tests on Individual Regression Coefficients and Groups of Coefficients

Tests on Groups of Coefficients

- The regression sum of squares for the reduced model is

$$SS_R(\beta_2) = \hat{\beta}'_2 X'_2 y \quad (p - r \text{ degrees of freedom})$$

- The regression sum of squares due to β_1 , given that β_2 is already in the model, with r degrees of freedom is $SS_R(\beta_1|\beta_2) = SS_R(\beta) - SS_R(\beta_2) = \hat{\beta}'_1 X'_1 y - \hat{\beta}'_2 X'_2 y$

- The test statistic used is F-value approach $\rightarrow$ partial F-test:

where

$$\hat{\sigma}^2 = \frac{y'y - \hat{\beta}'X'y}{n - p}$$

$$F_0 = \frac{SS_R(\beta_1|\beta_2)/r}{\hat{\sigma}^2}$$

- Reject H_0 , if $F_0 > F_{\alpha, r, n-p} \rightarrow$ at least one of the parameters in $\beta_1 \neq 0$.

Tests on Individual Regression Coefficients and Groups of Coefficients

Tests on Groups of Coefficients

- Compute $SS_R(\beta_j|\beta_i; i \neq j)$ to evaluate the contribution of an individual predictor.
- The increase in the regression sum of squares due to adding x_j to a model that already includes $x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_k$
- The partial F-test F_0 on a single variable x_j = the square of the t-test statistic t_0 .

Confidence Intervals on Individual Regression Coefficients

- Necessary to construct confidence interval (CI) estimates for the parameters (β) in a linear regression model.
- Procedure assumes that the model errors are normally and independently distributed with mean zero and variance σ^2 .
- The covariance matrix $V(\hat{\beta}) = \sigma^2 (X'X)^{-1}$.
- This CI can be written as

$$\hat{\beta}_j - t_{\alpha/2, n-p} se(\hat{\beta}_j) \leq \beta_j \leq \hat{\beta}_j + t_{\alpha/2, n-p} se(\hat{\beta}_j)$$

where

$$se(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 C_{jj}}$$

- This CI does not include zero $\rightarrow$ this is equivalent to rejecting (at $\alpha = 0.05$), the H_0 that the regression coefficient $\beta_1 = 0$.

Confidence Intervals on the Mean Response

- A confidence interval on the mean response at a particular combination of the predictor or regressor variables ($x_{01}, x_{02}, \dots, x_{0k}$) can be obtained.

- We know that the k predictors and an intercept term can be represented as x_0 ,

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1k} \\ 1 & x_{21} & x_{22} & \dots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nk} \end{bmatrix}$$

$$x_0 = \begin{bmatrix} 1 \\ x_{01} \\ \vdots \\ x_{0k} \end{bmatrix}$$

- The expected value (mean) of the response variable y given a specific value x_0 in a linear regression model is $E[y(x_0)] = \mu_{y|x_0} = x_0' \beta$.

- To compute the estimator, substitute $\hat{\beta} = \beta$

$$\hat{y}(x_0) = \hat{\mu}_{y|x_0} = x_0' \hat{\beta}$$

Confidence Intervals on the Mean Response

- With the assumption that $\hat{\beta}$ is normally distributed and an unbiased estimator of β , a $100(1 - \alpha)$ percent CI on the mean response at the point x_0 is

$$\hat{y}(x_0) - t_{\alpha/2, n-p} \sqrt{\hat{\sigma}^2 x_0' (X'X)^{-1} x_0} \leq \mu_{y|x_0} \leq \hat{y}(x_0) + t_{\alpha/2, n-p} \sqrt{\hat{\sigma}^2 x_0' (X'X)^{-1} x_0}$$

Critical value from the t-distribution with $(n-p)$ degrees of freedom

estimated variance of the errors (residual variance)

variance of the predicted mean response at x_0

standard error of the fitted response

X → **Design / model matrix** containing all predictor values for the regression model

(X'X)⁻¹ → **inverse of the information matrix**, used to measure the variability in the estimated coefficients

Confidence Intervals on the Mean Response

- α → The significance level (probability of making an error)
- $1-\alpha$ → The confidence level (probability that the interval contains the true value)
- $100(1 - \alpha)\%$: Expresses this probability as a percentage.
- 90% confidence interval ($\alpha=0.10$)
- 95% confidence interval ($\alpha=0.05$) → Most commonly used.
- 99% confidence interval ($\alpha=0.01$)
- The interval accounts for uncertainty in estimating the regression coefficients.
- A narrower CI indicates a more precise estimate
- A wider CI suggests greater uncertainty.

PREDICTION OF NEW OBSERVATIONS

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- A regression model can be used to predict future observations on the response y corresponding to a particular set of values of the predictor or regressor variables, say, $x_{01}, x_{02}, \dots, x_{0k}$.

- A point estimate of the future observation $y(x_0)$ at x_0 is

$$\hat{y}(x_0) = x_0' \hat{\beta}$$

$$x_0 = \begin{bmatrix} 1 \\ x_{01} \\ \vdots \\ x_{0k} \end{bmatrix}$$

- The prediction error in this case is

$$y(x_0) - \hat{y}(x_0)$$

- The variance of this prediction error is

$$\text{Var} [y(x_0) - \hat{y}(x_0)] = \text{Var} [y(x_0)] + \text{Var} [\hat{y}(x_0)] = \sigma^2 + \sigma^2 x_0' (X'X)^{-1} x_0 = \sigma^2 \left[1 + x_0' (X'X)^{-1} x_0 \right]$$

PREDICTION OF NEW OBSERVATIONS

- Estimate the error variance $\sigma^2 = \frac{\text{prediction error}}{\text{variance of this prediction error}}$

$$\frac{y(\mathbf{x}_0) - \hat{y}(\mathbf{x}_0)}{\sqrt{\hat{\sigma}^2 [1 + \mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0]}}$$

- The predicted interval is based on the probability with respect to the error variance and α with $(n-p)$ as degree of freedom

$$P \left(-t_{\alpha/2, n-p} \leq \frac{y(\mathbf{x}_0) - \hat{y}(\mathbf{x}_0)}{\sqrt{\hat{\sigma}^2 [1 + \mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0]}} \leq t_{\alpha/2, n-p} \right) = 1 - \alpha$$

- The $100(1 - \alpha)$ percent PI is

$$\begin{aligned} \hat{y}(\mathbf{x}_0) - t_{\alpha/2, n-p} \sqrt{\hat{\sigma}^2 [1 + \mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0]} &\leq y(\mathbf{x}_0) \\ &\leq \hat{y}(\mathbf{x}_0) + t_{\alpha/2, n-p} \sqrt{\hat{\sigma}^2 [1 + \mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0]} \end{aligned}$$

- PI > CI at the same point

PREDICTION OF NEW OBSERVATIONS

CONFIDENCE INTERVAL (CI)	PREDICTION INTERVAL(PI)
An interval estimate on the mean of the response distribution at a specific point.	An interval estimate on a single future observation from the response distribution at that point.